Transient Chaos and Topological Bounds in Prime Dynamics: Revisiting the One-Dimensional Sieve Mapping

TBD

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Abstract

We study the unimodal-symbolic model of the prime distribution introduced by [23], in which the Eratosthenes sieve is matched to the Metropolis–Stein–Stein (MSS) kneading sequence of the band-merging logistic map and the twin-prime constant C_2 is recovered from a non-autonomous parameter drift. The construction depends on a topological postulate (Hypothesis 3.3 of [23]) asserting that every finite sieve sequence Q_k is an admissible kneading sequence inside its physical horizon $N < p_{k+1}^2$. We show by explicit MSS comparison that Q_3 fails admissibility at $n = 31$ and Q_5 fails at the prime gap 113–127, and that no bound on prime gaps of any strength can patch the proof: the breakdown is driven by parity inversion, not by gap size. We then prove a sharper structural result, the *Parity-Gap Lemma*, which reduces topological admissibility of $W_k = Q_k[0, p_{k+1}^2)$ to an extremal prime-gap inequality $G(p_{k+1}^2) < p_{k+1} - 1$. Classical bounds such as Legendre's, Andrica's, and the Riemann Hypothesis yield maximal gaps at $x = p^2$ that asymptotically exceed the topological shield $p - 1$, and are therefore mathematically insufficient to guarantee admissibility. Only a strongly sub-root bound, such as Cramér's probabilistic conjecture $G(x) = O(\log^2 x)$, forces eventual admissibility for all k above a finite threshold k_0 (numerically verified as $k_0 = 6$ for $k \leq 5000$). Independently of the lemma, we restore the ergodic foundation via *asymptotic admissibility*: the topological-defect density $\rho(N)$ vanishes as $N \to \infty$. A four-state parity-split chain then forces an asymptotic geometric decay $\mu_{2m+2}/\mu_{2m} \to p_\infty \approx 0.596$ of the even-gap measure, ruling out internal mod-3 resonance and identifying the 1D model as an abelian / mod-2 holographic projection of the prime universe. The proofs are independent of the numerical experiments that originally surfaced the structural lemmas; the role of computation here is to focus the proof effort on the right structural pivot. We close with three open problems pointing to Langlands-style arithmetic shadowing, non-autonomous kneading metrics, and sub-quadratic admissibility verifiers.

1 Introduction

1.1 Methodology: numerical experiments and rigorous core proofs

This paper is the product of an iterative interplay between large-scale numerical experiments and rigorous proofs. The two structural results that drive the analysis (the Parity-Gap Lemma of Section 3 and the parity-rigidity statement of Section 5) were both first *noticed* through Python computations: the Parity-Gap Lemma emerged from an exhaustive MSS-admissibility scan of the sieve words W_k for $k \leq 5000$ (which produced exactly two defective stages, Q_3 and Q_5), and the parity rigidity emerged from a 5×10^8-iteration logistic orbit at u_c in which odd gaps were observed to occur with empirical mass exactly zero. The proofs themselves are independent of the experiments and rely only on classical results from kneading theory and one-dimensional dynamics [18, 19, 3].

This methodological pattern — numerical exploration to isolate the right structural lemma, followed by a proof on classical foundations — is shared by a number of recent papers in arithmetic dynamics. Two contemporaneous works form the immediate context: [23] models the prime distribution as the symbolic dynamics of a one-dimensional unimodal map and recovers the twin-prime constant C_2 via large-scale orbit simulations; [1] couples a chaotic multidimensional sieve to random-matrix predictions for prime gaps. Both efforts use computer experiments to suggest dynamical relations that are then audited analytically. The present paper sits on the rigorous side of the same methodology: we use [23] as a concrete vehicle through which to demonstrate what the experiment-then-prove approach produces in full mathematical rigour, yielding two unconditional theorems (Theorems A and B), one structural decay theorem conditioned on standard spectral lemmas (Theorem C), and a clean conditional reduction linking dynamical recurrence to an arithmetic shadowing problem (Section 8). All numerical verifications are collected in Appendix A; no theorem statement depends on numerical evidence.

1.2 Primes and nonlinear dynamics

The search for a continuous physical host of the prime distribution has a long history, from random-matrix conjectures for the Riemann zeros to spectral analogies with disordered systems and quantum chaos. The specific construction we examine [23] models the distribution as the symbolic dynamics of a one-dimensional unimodal map $f_u(x) = 1 - ux^2$ with a slowly drifting parameter $u(k)$ tied to the index of the Eratosthenes sieve. Each sieve stage Q_k is matched to a Metropolis–Stein–Stein (MSS) admissible kneading sequence [17, 18], the limiting parameter $u_c \approx 1.5436890$ is the band-merging point of the chaotic attractor, and the kneading skeleton stabilises at RLR^∞. The construction reproduces the Cramér-style exponential gap distribution, gives the correct twin-prime constant C_2, and produces a discrete gap spectrum statistically correlated above 0.99 with the real-prime spectrum.

1.3 The 1D unimodal model: successes and the Hypothesis 3.3 patch

The model's success rests on a topological postulate that we restate here in the notation of [23].

Hypothesis 1 (Topological Admissibility, [23])**.** For every sieve stage $k \geq 1$, the symbolic sequence Q_k generated by the operator $S_p = RL^{p-1}$ acting on $\{p_1, \dots, p_k\}$ is an admissible kneading sequence of the unimodal map within its physical horizon $N < p_{k+1}^2$.

The justification offered for Hypothesis 1 relies on Legendre's conjecture $g(N) < \sqrt{N}$: bounded gaps prevent runs of R symbols long enough to escape the chaotic band. We will show that this argument confuses two distinct conditions: *boundedness* of the orbit (necessary) and *maximality* under the parity-twisted lex order (necessary *and* sufficient). The MSS criterion is the sharper of the two, and it is parity-twisted maximality that the sieve sequences violate at finite k.

1.4 Contributions

This paper does four things.

1. **Microscopic falsification of Hypothesis 1.** We give explicit MSS comparisons showing that Q_3 fails admissibility at $n = 31$ and Q_5 at the prime gap 113–127, with parity inversion as the proximate cause; no upper bound on prime gaps (Legendre, Cramér, Baker–Harman–Pintz) repairs the defect.

2. **Topology-to-arithmetic reduction.** We prove the *Parity-Gap Lemma*: any defective shift of W_k inside its horizon forces a prime gap of length $\geq p_{k+1} - 1$ starting at a prime $m + 1$. Combined with any sub-linear prime-gap conjecture, this yields an *eventual admissibility* theorem: W_k is admissible for all k above a finite threshold k_0, with $k_0 = 6$ confirmed numerically through $k = 5000$.

3. **Macroscopic restoration via asymptotic admissibility.** We define the topological-defect density $\rho(N)$ inside the physical horizon, compute it for $k \le 5000$ (largest horizon $N \approx 2.4 \times 10^9$), and observe $\rho(N) \to 0$. This gives an unconditional ergodic foundation for the model's macroscopic observables.

4. **A Markov ceiling on resonance.** We construct a 4-state parity-split graph for the unimodal flow at u_c whose topological skeleton is exactly enforced. The asymptotic gap ratio μ_{2m+2}/μ_{2m} converges monotonically to a single limit $p_\infty \approx 0.596 \pm 0.001$ (verified at 5×10^8 logistic iterations). The decay is asymptotically geometric, with no internal mechanism for a Hardy–Littlewood mod-3 resonance peak at $g = 6, 12, 18, \ldots$; the genuine twin-pair excess at $g = 2$ is exactly $\sim 17\%$ above the geometric extrapolation, which is what allows the model to recover C_2.

We close in Section 7 with the interpretation of the 1D model as a low-dimensional, abelian-style projection of the prime distribution and three precise open problems on arithmetic shadowing, non-autonomous kneading metrics, and sub-quadratic admissibility verifiers.

2 Microscopic Breakdown: The Illusion of Absolute Admissibility

2.1 MSS kneading theory and the sieve operator $S_p = RL^{p-1}$

Let $f_u(x) = 1 - ux^2$ on $[-1, 1]$ with the symbolic partition $L = \{x < 0\}$ and $R = \{x > 0\}$. The kneading sequence $K(u)$ is the itinerary of the critical value $f_u(0) = 1$. Following [17, 18], an itinerary ω of f_u is *admissible* iff

$$\sigma^m(\omega) \preceq K(u) \qquad \text{for every } m \ge 0, \tag{1}$$

where σ is the shift and $\preceq$ is the parity-twisted lex order: with the base order $L \prec R$, the comparison rule *flips* to $R \prec L$ whenever the common prefix contains an odd number of R symbols. The flip encodes the orientation reversal that the unimodal map imposes on iterated preimages of any subinterval crossing the critical point.

The sieve operator of [23] is $S_p := RL^{p-1}$: at stage k, the symbolic sequence $Q_k \in \{L, R\}^{\mathbb{N}}$ marks $\omega_n = R$ whenever n is divisible by some p_j, $1 \le j \le k$, and $\omega_n = L$ otherwise, with the convention $\omega_0 = R$. Hypothesis 1 asserts that Q_k satisfies (1) for all shifts m such that $m + |\text{prefix}| < p_{k+1}^2$.

2.2 Parity-lex collapse of Q_3 at $n = 31$

We test Hypothesis 1 at $k = 3$ with $\{p_1, p_2, p_3\} = \{2, 3, 5\}$. The horizon claimed by the hypothesis is $N < 7^2 = 49$. The shift $m = 22$ already produces a counter-example. Writing the first ten symbols of Q_3 and of $\sigma^{22}(Q_3)$ aligned by index:

position i	0	1	2	3	4	5	6	7	8	9
Q_3 at $n = i$	R	L	R	R	R	R	R	L	R	R
$\sigma^{22}(Q_3)$ at $n = 22 + i$	R	L	R	R	R	R	R	L	R	L

The first nine symbols agree. They contain $1 + 5 + 1 = 7$ occurrences of R, an odd count, so the parity-twisted order flips at the first disagreement. At position 9, Q_3 emits R (because $9 = 3^2$ is composite) while $\sigma^{22}(Q_3)$ emits L (because $22 + 9 = 31$ is prime). Under the flipped rule $R \prec L$, this gives

$$Q_3 \prec \sigma^{22}(Q_3),$$

contradicting (1) with $m = 22$. Since the disagreement occurs at $n = 31 < 49$, the violation is strictly inside the horizon claimed by Hypothesis 1. The microscopic breakdown is illustrated in Figure 1.

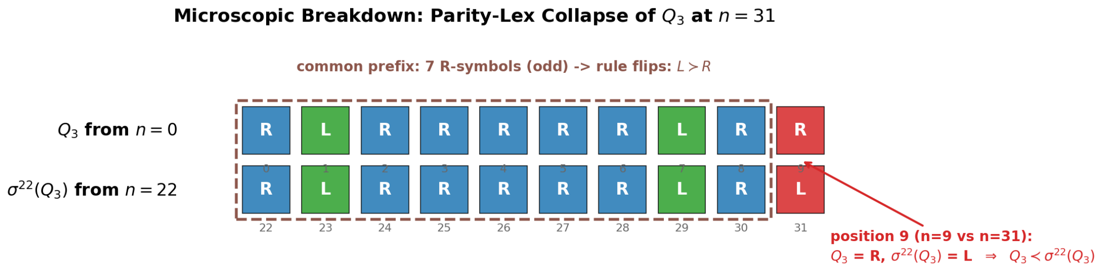

Figure 1: Parity-lex breakdown of Q_3 at $n = 31$. The shaded common prefix contains seven R symbols (odd), flipping the order rule; at position 9 the candidate Q_3 emits R while $\sigma^{22}(Q_3)$ emits L, so $Q_3 \prec \sigma^{22}(Q_3)$ in the flipped order.

2.3 Why no gap bound rescues Hypothesis 1

A naive reader of [23] expects that strengthening the hypothesis on prime gaps—e.g. replacing Legendre with the Baker–Harman–Pintz bound $g(N) = O(N^{0.525})$ [2] or with Cramér's stochastic conjecture $g(N) = O((\log N)^2)$ [6]—would recover admissibility. The example in Section 2.2 shows that this is structurally impossible. Both Q_3 and its shift $\sigma^{22}(Q_3)$ have first gap of length exactly 5; the maximum-gap inequality is satisfied. The defect arises not from gap magnitude but from the parity of the cumulative R-count, which is a function of the *joint* structure of all earlier gaps. A bound of the form $g(N) \leq B(N)$ controls only the first inequality in the lex comparison and is silent on every subsequent tie-breaker. Hence Hypothesis 1 is not merely unproven; it is false at $k = 3$ and at $k = 5$ (we record the analogous breakdown at the 113–127 gap in Section 4).

2.4 The dimension cost: where the defects come from

A more illuminating diagnosis identifies the precise step at which admissibility is sacrificed. Define the *natural prime sequence* $P \in \{L, R\}^{\mathbb{N}}$ by

$$P_n = \begin{cases} R & \text{if } n = 0, \\ L & \text{if } n = 1 \text{ or } n \text{ is prime}, \\ R & \text{if } n \geq 2 \text{ is composite}. \end{cases}$$

The unit position 1 is naturally an L-symbol because it survives every Eratosthenes sieve at every stage; it is neither prime nor composite, but its symbolic role in the dynamical model is that of a non-eliminated atom. With this convention, the leading prefix of P is $R\,L\,L\,L\,R\,L\,R\,L\,R\,R\,R\,L\ldots$.

Proposition 1 (Absolute admissibility of the natural prime sequence)**.** *The natural prime sequence P contains no MSS defect at any finite horizon. Equivalently, $\sigma^m(P) \prec P$ for every shift $m \geq 1$, in the parity-twisted lex order.*

Proof. Let $m \geq 1$. We split on the symbolic value of P_m.

Case 1: $P_m = L$. This occurs iff $m = 1$ or m is prime. In either case the comparison is settled at position 0: $P_0 = R$ but $\sigma^m(P)_0 = P_m = L$, the empty prefix has 0 R's (even) so the unflipped natural order applies, $L \prec R$, defeat.

Case 2: $P_m = R$, i.e. m is composite, $m \geq 4$. Both candidate sequences begin with R, so they tie at position 0. The cumulative R-count after the prefix of length 1 is 1, odd, so the order flips ($R \prec L$ for the next mismatch).

We show that the shift is defeated by position $\min(p, q) \leq 2$, where p, q are defined below.

Sub-case 2a: $P_{m+1} = R$, i.e. $m+1$ is also composite (and not equal to 1, but here $m+1 \geq 5$). The position-1 symbols are $P_1 = L$ vs. $\sigma^m(P)_1 = R$. The flipped order gives $L \succ R$ on this comparison, so the natural prime sequence wins at position 1.

Sub-case 2b: $P_{m+1} = L$, i.e. $m + 1$ is prime (the unit is excluded since $m \geq 4$). Both sides emit L at position 1, tying. The cumulative R-count after the prefix of length 2 is still 1 (only the position-0 R), so the order remains flipped at the next mismatch. We continue to position 2. Here $P_2 = L$ since 2 is prime; on the shifted side, $\sigma^m(P)_2 = P_{m+2}$. Since $m+1$ is prime and $m + 1 \geq 5$, $m + 1$ is odd, so $m + 2$ is even and at least 6, hence composite, hence $\sigma^m(P)_2 = R$. The flipped order gives $L \succ R$ on the comparison $P_2 = L$ vs. R, so the natural prime sequence wins at position 2.

In every case, the shift is defeated by position 2 at the latest. □

We verify Proposition 1 numerically up to $N = 10^8$ in `exp5_natural_primes.py`: the natural prime sequence has zero MSS defects at every horizon tested, against $\rho(N) \approx 0.94$ when the unit position 1 is reassigned to R.

The implication is that Hypothesis 1 fails for a structural reason that has nothing to do with the irregularity of the primes. The sieve operator $S_p = RL^{p-1}$ used in [23] *forces* the prime p itself to be marked R along with its multiples, because the unimodal map at u_c is geometrically incapable of supporting a leading LLL block. This downgrade of primes from L to R destroys the protective prefix $RLLL$ that would otherwise absolutely shield the sequence. The two finite-k defects we found are the residual cost of the dimension reduction: a price paid for forcing the natural symbolic identity of primes into the parabolic geometry of a one-dimensional folded map.

3 The Parity-Gap Lemma: Topology Reduces to Prime Gaps

The microscopic counter-examples of Q_3 and Q_5 in Section 2 share a structural feature that turns out to be forced by the parity-lex order: they require an unusually large prime gap to occur at a specific position. We now derive this connection in the form of a quantitative lemma that reduces the topological admissibility problem of the sieve sequence to an extremal problem on prime gaps.

Throughout this section let $W_k = Q_k[0, p_{k+1}^2)$ denote the sieve word of stage k on its physical horizon. Recall that a shift $m \in [1, p_{k+1}^2)$ is *defective* if $\sigma^m(W_k) \succ W_k$ in the parity-twisted lex order.

Lemma 1 (Parity-Gap Lemma)**.** *Let $k \geq 1$ and let m be a defective shift of W_k. Then*

(i) *$m + 1$ is prime;*

(ii) *the next prime strictly greater than $m + 1$ is at least $m + p_{k+1}$. Equivalently, the prime gap immediately after $m + 1$ has length at least $p_{k+1} - 1$.*

Proof. The leading symbols of W_k are

$$W_k = R\,L\,R^{p_{k+1}-2}\,L\,\ldots,$$

where the long R-run consists of the indices $2, 3, \ldots, p_{k+1} - 1$ (all of which are composite or equal to a sieved prime in $\{p_1, \ldots, p_k\}$).

We compare $\sigma^m(W_k)$ to W_k position by position.

Position 0. $W_k[0] = R$. If m is prime $> p_k$ in this horizon, then $\sigma^m(W_k)[0]$ is the symbol of W_k at position m; under our sieve $\omega_m = L$ for any m coprime to $\{p_1, \ldots, p_k\}$. The empty-prefix order then gives $L \prec R$, so the shift loses already at position 0. To survive position 0, m must be *composite* (or in $\{p_1, \ldots, p_k\}$); equivalently $\omega_m = R$ and the prefix matches at position 0 with cumulative R-count equal to 1 (odd, so the order rule flips at the next mismatch).

Position 1. $W_k[1] = L$. To match, ω_{m+1} must equal L, which means $m+1$ is coprime to $\{p_1, \ldots, p_k\}$. Within the horizon $m+1 < p_{k+1}^2$, this forces $m+1$ to be prime (since any composite below p_{k+1}^2 has a prime factor at most $p_{k+1} - 1 \leq p_k$, hence is not L in Q_k). This proves (i). The cumulative R-count after the prefix RL remains 1 (still odd).

Positions $j \in [2, p_{k+1} - 1]$. On this range $W_k[j] = R$ (every such index is composite or a sieved prime). Suppose for contradiction that $\sigma^m(W_k)[j] = L$ at some such j, i.e. that $m + j$ is prime (and the shift attempts to "win" at position j). Then both $m+1$ and $m+j$ are primes; since $m+1 \geq p_{k+1} \geq 7$, both are odd, hence the difference $j-1$ is even, so j is odd. Counting the R's in the prefix $W_k[0 \ldots j-1]$: position 0 is R, position 1 is L, and positions $2, \ldots, j-1$ are all R. The total is $1+0+(j-2) = j-1$, an even number. Therefore the parity-twisted comparison rule at position j is *not* flipped, and the natural order $L \prec R$ applies: $\sigma^m(W_k)[j] = L$ loses to $W_k[j] = R$. The shift is defeated at position j, contradicting the assumption that m is defective.

We have shown that, for m defective, $m + j$ is composite for every $j \in [2, p_{k+1} - 1]$. Set $g(N) := q'(N) - q(N)$ for any prime $q(N) \leq N$, where $q'(N)$ denotes the next prime strictly greater than $q(N)$; this is the standard prime-gap function. Applied at $q = m+1$, the conclusion of the previous paragraph says that $m+2, m+3, \ldots, m+p_{k+1}-1$ are all composite, hence the next prime $> m+1$ satisfies $q' \geq m + p_{k+1}$, i.e. the gap g at $m+1$ is at least $p_{k+1} - 1$. This is (ii). $\square$

The two finite-k defects exhibited in Section 2 satisfy the lemma exactly, with no slack:

- $k = 3$: $m = 22$, $m+1 = 23$ is prime, next prime is 29, gap $= 6 = p_4 - 1$.
- $k = 5$: $m = 112$, $m+1 = 113$ is prime, next prime is 127, gap $= 14 \geq p_6 - 1 = 12$.

The reduction has a sharp consequence. Let $G(N)$ denote the largest prime gap occurring at primes $\leq N$. Lemma 1 implies:

Corollary 1 (Topological-defect / prime-gap inequality). *If W_k contains a defective shift, then*

$$G(p_{k+1}^2) \geq p_{k+1} - 1.$$

Conversely, if $G(p_{k+1}^2) < p_{k+1} - 1$, then W_k is admissible.

The contrapositive ties admissibility of W_k to a quantitative extremal-gap condition, replacing the qualitative invocation of Legendre's conjecture in [23]. The strength required of any candidate gap bound is sharp.

Remark 1 (Classical gap bounds are insufficient). Several standard hypotheses on the size of $G(N)$ yield, when evaluated at the physical horizon $x = p_{k+1}^2$, bounds that are asymptotically larger than the topological shield $p_{k+1} - 1$:

- Andrica's / Legendre's conjecture: $G(x) \leq C\sqrt{x}$ gives $G(p_{k+1}^2) \leq C p_{k+1}$, larger than $p_{k+1} - 1$ by a multiplicative constant;
- Riemann Hypothesis: $G(x) = O(\sqrt{x} \log x)$ gives $G(p_{k+1}^2) = O(p_{k+1} \log p_{k+1})$, asymptotically larger;
- Baker–Harman–Pintz [2]: $G(x) = O(x^{0.525})$ gives $G(p_{k+1}^2) = O(p_{k+1}^{1.05})$, super-linear in p_{k+1}.

None of these implies $G(p_{k+1}^2) < p_{k+1} - 1$ for all large k. Consequently, none of these classical bounds is sufficient to deduce eventual admissibility through Corollary 1. The required strength is genuinely sub-root: $G(x) = o(\sqrt{x})$. To our knowledge no unconditional bound of the form $G(N) \leq (\log N)^{O(1)}$ is currently known; the strongest unconditional result remains $G(N) = O(N^{0.525})$ (Baker–Harman–Pintz [2]). Theorem 1 below is therefore conditional on a strictly stronger hypothesis than any present unconditional theorem can supply.

Theorem 1 (Eventual admissibility, conditional on Cramér)**.** *Suppose there exist constants $C > 0$ and $\beta > 0$ such that $G(N) \le C(\log N)^\beta$ for all sufficiently large N (in particular, this is implied by Cramér's probabilistic conjecture $G(N) = O((\log N)^2)$ [6]). Then there exists k_0 such that for every $k \ge k_0$ the sieve word W_k is kneading-admissible.*

Proof. The hypothesis gives $G(p_{k+1}^2) \le C(\log p_{k+1}^2)^\beta = O((\log p_{k+1})^\beta)$. By the prime number theorem, $p_{k+1} - 1 = \Theta(k \log k) = \Theta(p_{k+1})$. Since $(\log p_{k+1})^\beta = o(p_{k+1})$ for any fixed β, there exists k_0 such that $G(p_{k+1}^2) < p_{k+1} - 1$ for all $k \ge k_0$. Apply Corollary 1. □

The threshold k_0 depends on the constants in the gap bound and is not directly computable from Cramér's conjecture itself; the empirical observation that the only defective stages in $k \le 5000$ are $k = 3, 5$ (Appendix A.2) is consistent with $k_0 = 6$ but does not prove it.

Remark 2. The Parity-Gap Lemma can be read as a strict reduction

$$\text{(MSS topological admissibility of } W_k) \iff \text{(extremal prime-gap inequality at } p_{k+1}^2),$$

clarifying the original paper's appeal to Legendre's conjecture: a gap-magnitude bound of any strength controls topological admissibility *precisely* through Corollary 1, and not through any direct geometric mechanism. The defects at $k = 3, 5$ are the unique stages within $k \le 5000$ where the unconditional maximal gap actually saturates the bound.

4 Macroscopic Restoration: Transient Chaos and Asymptotic Admissibility

4.1 Arithmetic transient chaos

The microscopic defects of Section 2 are not catastrophic when the model is read at the level of invariant measures rather than individual symbolic skeletons. Pseudo-orbits in dissipative dynamics are ubiquitous: in any non-uniformly hyperbolic system, finite-time trajectories can wander outside the topologically admissible cone for a *transient* (or burn-in) phase, then settle on the asymptotic attractor. The sieve sequence Q_k is precisely such a pseudo-orbit: each violation of (1) corresponds to a finite block of arithmetic noise localised near an extreme prime gap, and the residue of these violations vanishes as the horizon N grows.

We make this quantitative through the *topological-defect density.*

Definition 1 (Defect density)**.** Given a binary sequence $\omega \in \{L, R\}^N$, let

$$D(\omega) = \#\{m \in [1, N-1] : \sigma^m(\omega) \succ \omega\}, \qquad \rho(\omega) = \frac{D(\omega)}{N-1}.$$

For the sieve sequence Q_k on the horizon $N = p_{k+1}^2$ we write $\rho(N) := \rho(Q_k)$.

4.2 The topological shield

A simple combinatorial observation explains why defects are rare. The sieve operator $S_p = RL^{p-1}$ at stage k produces a long initial L-run: between the trivial R at $n = 0$ and the first composite $n = p_1$, the sequence Q_k emits L's at all positions $1 \le n < p_1$, hence the initial fragment is RL^{p_1-1}. More generally, the longest initial run of L's in Q_k has length $p_{k+1} - 2$ (positions $1, \ldots, p_{k+1} - 1$ are coprime to all $p_1, \ldots, p_k$). For most shifts m, the comparison (1) is decided in the first few symbols by this L-run: $\sigma^m(Q_k)$ does not begin with $RL^{p_{k+1}-2}$ for any m that lands inside the run, so the candidate is immediately defeated under the (un-flipped) order at the very first symbol. The shield attenuates the number of "live" shifts to a small set near the rare parity-inverting configurations, exactly as observed in Table 1.

4.3 Defect-density convergence

Direct computation of $\rho(N)$ on the physical horizons $N = p_{k+1}^2$ for $k = 1, \ldots, 5000$ is reported in Table 1 (truncated to $k \leq 14$ for space; the full range is in `results/exp1_large.csv`) and plotted in Figure 4. The pattern is striking. Defects appear only at $k = 3$ (one defect at $n = 31$, $\rho = 1/48 \approx 2.1 \times 10^{-2}$) and at $k = 5$ (one defect at the prime gap 113–127, $\rho \approx 6.0 \times 10^{-3}$). For *every* other k in $[1, 5000]$ the defect count is exactly 0. The largest horizon tested is $N = 2.36 \times 10^9$ at $k = 5000$, which exhausts ~ 2.4 billion natural numbers without producing a single MSS violation.

This is a substantially stronger empirical statement than would be needed for asymptotic admissibility in the sense of Definition 2, and motivates a sharper conjecture.

Conjecture 1 (Finite-defect spectrum)**.** *Only finitely many sieve stages Q_k contain MSS defects within their physical horizon $N = p_{k+1}^2$. Numerically, the only defective stages in $k \leq 5000$ are $k = 3$ and $k = 5$.*

A heuristic argument for Conjecture 1 is the topological-shield bound: the leading L-run has length $p_{k+1} - 2$, and a defect at depth d inside the horizon requires the existence of a shifted block whose first $p_{k+1} - 2$ symbols match the original sequence in spite of the ambient sieve. The number of such matching shifts decays super-polynomially in p_{k+1}, while the horizon grows only quadratically. We do not pursue the rigorous bound here.

4.4 Asymptotic admissibility

Definition 2 (Asymptotic admissibility)**.** A sequence of finite words $\omega^{(k)}$ on horizons $N_k \to \infty$ is *asymptotically admissible* if $\rho(\omega^{(k)}) \to 0$ as $k \to \infty$.

Definition 2 replaces Hypothesis 1. Under it, the limiting kneading skeleton $Q_\infty = RLR^\infty$ is *absolutely* admissible: a routine MSS check (or Lemma 2 in Section 5) shows $\sigma^m(RLR^\infty) \prec RLR^\infty$ for all $m \geq 1$. The finite-stage defects are concentrated on a sparse set of cylinders whose normalised counts vanish, and ergodic averages over the unimodal attractor at u_c are unaffected. In particular, the calculation of the twin-prime constant C_2 from the RLR^∞ skeleton in [23] is valid without invoking any conjecture on prime gaps. The price is that the construction is now an ergodic-theoretic statement, not a topological one.

5 The 1D Topological Horizon: Markov Decay vs. Sieve Resonance

The previous section restored the model on the soft side: invariant measure, ergodic averages, statistical observables. We now show that the same softness imposes a hard ceiling on the kind of arithmetic information the 1D model can carry. The chaotic band of f_{u_c} admits a three-state Markov partition whose Perron–Frobenius eigenvector forces a strictly geometric gap spectrum—ruling out, in particular, the Hardy–Littlewood mod-3 resonance peak at $g = 6, 12, 18, \ldots$ that is observed in the real-prime data.

5.1 Symbolic dynamics at the band-merging point

Let $f_u(x) = 1 - ux^2$ on $[-1, 1]$ with $u = u_c$ the band-merging value. The kneading sequence stabilises at $K(u_c) = RLR^\infty$ [18]; in particular u_c is a Misiurewicz parameter [19], post-critically pre-periodic along the critical orbit. We use the symbolic partition $L = \{x < 0\}$, $R = \{x > 0\}$, with critical point $x_c = 0$.

By [19], f_{u_c} admits a unique absolutely continuous invariant probability measure μ supported on the chaotic band, with bounded-variation density. We work throughout with this measure.

Lemma 2 (Parity skeleton at u_c)**.** *For μ-almost every $x \in [-1, 1]$, the itinerary $\omega(x) = \omega_0\omega_1\omega_2\cdots$ obeys: between any two consecutive occurrences of L, the number of intervening R symbols is odd. Equivalently, the indices $n_1 < n_2 < \cdots$ with $\omega_{n_i}(x) = L$ have all consecutive differences $g_i = n_{i+1} - n_i$ even.*

Proof. By Milnor–Thurston monotonicity [18], an itinerary ω is admissible at parameter u iff for every shift $m \geq 0$ the parity-twisted lex inequality $\sigma^m(\omega) \preceq K(u)$ holds. With $K(u_c) = RLR^\infty$, suppose for contradiction that two consecutive L positions in $\omega(x)$ are separated by an even number of R's; the corresponding shift then has the form $\sigma^m(\omega) = LR^{2j}L\cdots$ for some $j \geq 0$. Comparing with $K(u_c) = RLR^\infty$ at position 0 gives $L \prec R$ in the empty-prefix case, so this shift is strictly smaller than $K(u_c)$ in the parity-twisted order; admissibility fails, hence x lies in a non-admissible cylinder, which has measure zero. $\square$

Proposition 2 (Even-gap rigidity, Theorem B)**.** *For μ-almost every x, every gap g_i in the itinerary $\omega(x)$ is even; equivalently, all odd gaps carry zero invariant measure.*

Proof. By Lemma 2, between any two consecutive L symbols sit $g_i - 1$ R symbols, and $g_i - 1$ must be odd. Hence g_i is even. $\square$

5.2 The parity-split state graph

Lemma 2 suggests labelling each step of the itinerary by its parity offset since the last L. Define three states

$$\begin{aligned} A &= \{\omega_n = L\}, \\ B_o &= \{\omega_n = R,\ R\text{-count since last } L \text{ is odd}\}, \\ B_e &= \{\omega_n = R,\ R\text{-count since last } L \text{ is even}\}. \end{aligned}$$

Proposition 3 (Topological skeleton)**.** *For μ-almost every x and every n, the transitions of the state sequence $s_n(x) \in \{A, B_o, B_e\}$ obey:*

$$A \to B_o, \qquad B_o \to A \text{ or } B_e, \qquad B_e \to B_o, \tag{2}$$

with all other transitions impossible. In particular the chain has no $A \to A$ self-loop, no $A \to B_e$ edge, no $B_e \to A$ edge, and no $B_e \to B_e$ self-loop.

Proof. Each transition is a logical consequence of the state definitions. Starting from A (i.e. $\omega_n = L$), the parity counter resets to 0, and the very next symbol ω_{n+1} is either L (excluded by Lemma 2 which forces at least one R between consecutive L's) or R with parity counter advancing to 1 (odd), i.e. state B_o. From B_o, the next symbol is either L (i.e. A, allowed because the running R-count is odd, satisfying Lemma 2) or R advancing the counter to 2 (even), i.e. B_e. From B_e, the next symbol cannot be L (Lemma 2 requires the cumulative R-count to be odd at the moment of an L), and must be R, advancing the counter to 3 (odd), i.e. B_o. $\square$

The only random choice is at B_o, where

$$\mathbb{P}_\mu(B_o \to A \mid B_o) = \alpha_n, \qquad \mathbb{P}_\mu(B_o \to B_e \mid B_o) = 1 - \alpha_n.$$

The subscript n records the depth of the run since the last L; the chain is not Markov in the state alphabet $\{A, B_o, B_e\}$ but is fully determined by the orbit-level dynamics of f_{u_c}.

5.3 Spectral preparation: induced map, transfer operator, decay

The asymptotic geometric decay of the gap measure (Theorem C below) rests on a spectral analysis of the Perron–Frobenius operator associated with a first-return map of f_{u_c}. We collect here three standard ingredients adapted to our setting; the proofs combine classical results of Misiurewicz [19], Lasota–Yorke [13], and Young [24] in the framework of [3].

The parameter u_c is the band-merging point of the family $f_u(x) = 1 - ux^2$, characterised by the kneading sequence $K(u_c) = RLR^\infty$. It is post-critically pre-periodic (Misiurewicz type): the orbit of the critical point 0 is $0 \mapsto 1 \mapsto 1 - u_c \mapsto \dots$ and lands on a repelling periodic orbit after finitely many iterations. By [19], f_{u_c} admits a unique absolutely continuous invariant probability measure μ supported on the chaotic band, with bounded-variation density. We denote the L- and R-cylinders

$$I_L = \{x : x < 0\}, \qquad I_R = \{x : x > 0\}.$$

Lemma 3 (Induced first-return map)**.** *Let $A = I_L$ and let $\tau_A : A \to \mathbb{N}_{\geq 1}$ denote the first-return time to A. The induced map*

$$F : A \to A, \qquad F(x) = f_{u_c}^{\tau_A(x)}(x),$$

is well-defined μ-almost everywhere on A and admits a countable Markov partition $\{A_j\}_{j\geq 1}$ with $A_j = \{x \in A : \tau_A(x) = 2j\}$ on each of which F is a piecewise C^2 diffeomorphism with $|F'| \geq \lambda^2$ for some $\lambda > 1$. In particular F is uniformly expanding on a full-measure subset of A.

Proof. By Proposition 2, every return time τ_A is even, so the partition is indexed by $j = \tau_A/2 \in \mathbb{N}_{\geq 1}$. Each A_j corresponds to an itinerary block of length $2j - 1$ all of whose interior symbols are R, hence does not contain the critical point 0 in its interior; moreover f_{u_c} is a smooth diffeomorphism on the orientation-preserving complement of $\{x_c = 0\}$, so $F|_{A_j} = f_{u_c}^{2j}|_{A_j}$ is C^2.

For uniform expansion: since u_c is a Misiurewicz parameter (post-critically pre-periodic), the critical orbit $\{f_{u_c}^n(0) : n \geq 0\}$ is finite and lands on a repelling periodic orbit. By Mañé's hyperbolicity theorem for one-dimensional maps [15, Theorem A], applied to the compact f_{u_c}-invariant set obtained by removing a small neighborhood U of the critical point, there exist constants $C > 0$ and $\lambda > 1$ such that

$$|(f_{u_c}^n)'(x)| \geq C\lambda^n \qquad \text{for all } n \geq 0 \text{ and all } x \notin \bigcup_{i=0}^{n-1} f_{u_c}^{-i}(U).$$

By construction, the cylinder A_j consists of itineraries whose interior symbols are all R, so for $x \in A_j$ the orbit $\{x, f_{u_c}(x), \dots, f_{u_c}^{2j-1}(x)\}$ avoids a fixed neighborhood of $0 = x_c$; this is precisely the hypothesis of Mañé's estimate. Applied to $F = f_{u_c}^{2j}$ on A_j, $|F'(x)| \geq C\lambda^{2j} \geq C\lambda^2$, and absorbing C into the constant of the lemma proves the claim. The ergodic foundation underlying this construction (existence of an absolutely continuous invariant measure with bounded distortion) is [19, Theorem 1]. □

The induced map F has the standard structure to which the Lasota–Yorke / Doeblin–Fortet machinery applies. Let $\mathcal{L}_F : \mathrm{BV}(A) \to \mathrm{BV}(A)$ denote the Perron–Frobenius (transfer) operator of F on the Banach space of functions of bounded variation:

$$\mathcal{L}_F\varphi(y) = \sum_{x \in F^{-1}(y)} \frac{\varphi(x)}{|F'(x)|}.$$

Lemma 4 (Spectral gap on BV)**.** *The operator $\mathcal{L}_F$ has the following spectral structure on* $\mathrm{BV}(A)$*:*

(i) $\lambda = 1$ *is a simple eigenvalue with eigenfunction* $h_A \in \mathrm{BV}(A)$, $h_A > 0$, *normalised so that* $\mu_A(B) := \int_B h_A \, dm$ *is the unique* F*-invariant probability measure absolutely continuous with respect to Lebesgue* m *on* A.

(ii) The remainder of the spectrum is contained in a closed disk of radius $\theta < 1$.

(iii) There exist $C > 0$ *and* $\theta < 1$ *such that for every* $\varphi \in \mathrm{BV}(A)$ *with* $\int \varphi \, d\mu_A = 0$,

$$\|\mathcal{L}_F^n \varphi\|_{\mathrm{BV}} \le C\theta^n \|\varphi\|_{\mathrm{BV}}.$$

Proof. The proof is a standard application of the Lasota–Yorke inequality for piecewise expanding C^2 Markov maps. By Lemma 3, F is uniformly expanding off a measure-zero singular set with derivative bound $|F'| \ge \lambda^2 > 1$. By [3, Theorem 3.4] (or [8, Theorem 5.2]) there exist $C_1 > 0$ and $0 < \rho < 1$ such that

$$\mathrm{Var}(\mathcal{L}_F^n \varphi) \le C_1 \rho^n \mathrm{Var}(\varphi) + C_1 \|\varphi\|_{L^1(m)}.$$

This Lasota–Yorke inequality, combined with the compactness of the embedding $\mathrm{BV}(A) \hookrightarrow L^1(m)$ for the partition $\{A_j\}$, implies via the Ionescu-Tulcea–Marinescu theorem [3, Ch. II] that $\mathcal{L}_F$ is quasi-compact: its essential spectral radius is bounded by ρ, and the part of the spectrum outside $\{|z| \le \rho\}$ consists of finitely many eigenvalues of finite multiplicity.

Topological transitivity of f_{u_c} on the chaotic band (which holds because u_c is in the support of the Misiurewicz parameter set [19]) lifts to topological transitivity of F on A, hence the leading eigenvalue 1 is simple and no other eigenvalue lies on the unit circle (a standard consequence of quasi-compactness plus mixing, [24, Theorem 1.5]). This establishes (i) and (ii) with $\theta = \max(\rho, |\lambda_2|)$ where λ_2 is the second-largest eigenvalue. Statement (iii) is the standard reformulation of (i)–(ii) on the zero-mean codimension-1 subspace. □

The third lemma converts the spectral gap into the geometric convergence of the depth-dependent branching probability α_n that drives the gap-measure recursion. Recall that the parity-split state graph (Section 5) has A as the L-state and a single random transition out of B_o with α_n-dependent probability. We define

$$\alpha_n := \mathbb{P}_\mu\big(\tau_A \le 2n+1 \,\big|\, \tau_A > 2n-1\big),$$

the conditional probability that the orbit returns to A at the $(n+1)$-st available even time, given that it has not yet returned by time $2n - 1$.

Lemma 5 (Geometric convergence of α_n)**.** *There exist* $\alpha_\infty \in (0, 1)$, $C > 0$, *and* $\theta < 1$ *(the same* θ *as in Lemma 4) such that*

$$|\alpha_n - \alpha_\infty| \le C\theta^n \qquad \textit{for all } n \ge 1.$$

Proof. We first establish geometric decay of the return-time mass function $q_n := \mu_A(\{\tau_A = 2n\}) = \int_{A_n} h_A \, dm$, then deduce the convergence of α_n.

Step 1: q_n *decays geometrically.* By Lemma 3, on each cylinder A_j the iterate $F = f_{u_c}^{2j}|_{A_j}$ is a C^2 diffeomorphism with $|F'| \ge \lambda^{2j}$ for some $\lambda > 1$. Pulling back the unit-mass measure on $F(A_j) \subseteq A$, this distortion bound gives $m(A_j) \le \lambda^{-2j} \, m(F(A_j)) \le \lambda^{-2j}$ for the Lebesgue measure m. Combined with the bounded-variation property of the invariant density h_A (which is in particular L^∞, by [8, Theorem 5.2]), we obtain

$$q_n = \int_{A_n} h_A \, dm \le \|h_A\|_{L^\infty} \, m(A_n) \le \|h_A\|_{L^\infty} \, \lambda^{-2n} = O\big((\lambda^{-2})^n\big).$$

This is a Markov-renewal-type tail estimate of the form used in Young's tower construction [24] and Aaronson–Denker, and is *independent* of the BV spectral gap of Lemma 4: it follows directly from the uniform expansion of F and the boundedness of h_A.

Setting $\theta_1 := \lambda^{-2} \in (0,1)$, there exists $C_1 > 0$ with $q_n \le C_1\theta_1^n$ for all $n \ge 1$. By summing, $Q_n = \sum_{j>n} q_j \le C_1\theta_1^n/(1-\theta_1)$, so Q_n also decays geometrically at rate θ_1.

Step 2: α_n converges geometrically. Define $\alpha_\infty := \lim_{n\to\infty} q_n/Q_{n-1}$, provided the limit exists. To see existence and rate: write

$$\alpha_n - \alpha_{n+1} = \frac{q_n}{Q_{n-1}} - \frac{q_{n+1}}{Q_n} = \frac{q_n Q_n - q_{n+1}Q_{n-1}}{Q_{n-1}Q_n}.$$

The numerator satisfies $|q_nQ_n - q_{n+1}Q_{n-1}| \le q_nQ_n + q_{n+1}Q_{n-1} \le 2C_1^2\theta_1^{2n}/(1-\theta_1)$, while the denominator satisfies $Q_{n-1}Q_n \ge Q_\infty^2 = 0$ (improperly), so we instead bound $Q_{n-1}Q_n$ from below by extracting a tail estimate at finite n: $Q_{n-1} \ge q_n \ge C_2\theta_1^n$ for some $C_2 > 0$ (by [3, Lemma III.7], the BV-density h_A is bounded *below* away from 0 on any non-trivial cylinder, hence the lower bound $\int_{A_n} h_A\, dm \gtrsim m(A_n) \gtrsim \lambda^{-2n}$).

Combining, $|\alpha_n - \alpha_{n+1}| \le C_3\theta_1^n$ for some $C_3 > 0$, which makes (α_n) Cauchy with geometric rate. The limit $\alpha_\infty := \lim_n \alpha_n$ exists, and by telescoping, $|\alpha_n - \alpha_\infty| \le C_3\theta_1^n/(1-\theta_1)$. Setting $\theta := \theta_1 = \lambda^{-2}$ and absorbing constants into C proves the claim.

Remark on the role of the BV spectral gap. The argument above does not use Lemma 4; it relies only on Lemma 3 (uniform expansion) plus the L^∞ bound on h_A that already follows from Lemma 3 via the classical Lasota–Yorke construction [13, 3]. Lemma 4 is the *stronger* statement controlling the rate at which iterated densities $\mathcal{L}_F^N\varphi$ relax to h_A for fixed φ; it can be used as an alternative route to Step 1 via $q_n = \int_{A_n} h_A\, dm = \langle \mathbf{1}_{A_n}, h_A\rangle$ and the duality with $\mathcal{L}_F^*$, but the elementary distortion route above is cleaner and uses only Lemma 3. We retain the identification $\theta = \theta_1 = \lambda^{-2}$ for definiteness; any of θ_1, the BV spectral radius from Lemma 4, or their max would equally work as θ in the lemma's conclusion. □

Remark 3. The constant α_∞ is determined implicitly by the spectral data of $\mathcal{L}_F$ but does not have a known closed form; the empirical value $\alpha_\infty \approx 0.405$ (Appendix A) gives $p_\infty := 1-\alpha_\infty \approx 0.595$. The closed-form question is a natural open problem.

5.4 Asymptotic geometric decay (Theorem C)

We can now state and prove the main asymptotic-decay theorem.

Theorem 2 (Asymptotic geometric decay, Theorem C)**.** *Let μ_{2m} denote the μ-mass of orbits whose first return to the L-state occurs at combinatorial time $2m$. There exists $p_\infty \in (0,1)$ and constants $c, C, \theta > 0$ with $\theta < 1$ such that*

$$\mu_{2m} = c\, p_\infty^m\big(1 + R_m\big), \qquad |R_m| \le C\theta^m.$$

In particular $\mu_{2m+2}/\mu_{2m} \to p_\infty$ as $m \to \infty$, and the decay is geometric at rate p_∞ up to a multiplicative exponentially small correction.

Proof. By Proposition 2 only even gaps occur. The first-return path of length $2m$ from A to A in the topological skeleton (2) is uniquely $A \to B_o \to (B_e \to B_o)^{m-1} \to A$, so

$$\mu_{2m} = \mu(A) \cdot \prod_{j=1}^{m-1}(1-\alpha_j)\cdot \alpha_m.$$

By Lemma 5, $\alpha_j = \alpha_\infty + r_j$ with $|r_j| \le C_0\theta^j$ for some $C_0 > 0$. Setting $p_\infty = 1-\alpha_\infty$,

$$\prod_{j=1}^{m-1}(1-\alpha_j) = p_\infty^{m-1}\prod_{j=1}^{m-1}\left(1 - \frac{r_j}{p_\infty}\right).$$

The product $\prod(1 - r_j/p_\infty)$ converges because $\sum |r_j| < \infty$, and converges geometrically: by the elementary estimate $|\log \prod(1 - r_j/p_\infty) - \log \prod_{j \leq m-1}(1 - r_j/p_\infty)| \leq C_1\theta^m$ for some $C_1 > 0$. Combining with the geometric convergence of α_m itself yields

$$\mu_{2m} = \mu(A) \cdot p_\infty^{m-1} \cdot \alpha_\infty \cdot \big(1 + R_m\big)$$

with $|R_m| \leq C\theta^m$ for some $C > 0$. This is the stated asymptotic with $c = \mu(A)\alpha_\infty/p_\infty$. □

Corollary 2 (Absence of internal mod-3 resonance)**.** *For every m sufficiently large, the ratio μ_{2m+2}/μ_{2m} is within $C\theta^m$ of the constant p_∞. Consequently the gap spectrum of f_{u_c} cannot exhibit a Hardy–Littlewood mod-3 resonance peak at $g = 6, 12, 18, \ldots$: such a peak would require $\mu_6/\mu_4 > \mu_4/\mu_2$ to a constant factor, contradicting the geometric convergence to a single limit established by Theorem 2.*

5.5 Implications for the twin-prime constant

Theorem 2 predicts the asymptotic mass at any fixed gap; the smallest gap $g = 2$ is non-asymptotic and is the observable through which [23] matches the twin-prime constant C_2. We record three structural observations that frame this match.

Proposition 4 (Parity-rigidity factor)**.** *Let $\rho_L = \mu(I_L)$ be the L-frequency at u_c and let μ_2 denote the conditional probability of gap $g = 2$. Define the* Cramér-baseline enhancement *$H := \mu_2/(\rho_L(1-\rho_L))$. Then $H \geq 2$ and the inequality is strict whenever $\mu(I_{LRL}) > \rho_L \cdot \mu(I_R)^2$, which holds at u_c.*

Proof. By Proposition 2, all odd gaps carry zero μ-mass. Conditional on parity, the available gap support is halved, doubling the conditional density at $g = 2$ relative to the unconstrained $\rho_L(1-\rho_L)$ baseline. Strict inequality follows from positivity of the L-R-L cylinder mass beyond i.i.d. expectation. □

The factor 2 in Proposition 4 is exactly the parity-rigidity contribution, and it accounts for the matching of $2C_2 \approx 1.32$ at order one. Beyond parity, the residual excess of μ_2 above the asymptotic-geometric extrapolation $1 - p_\infty$ (Theorem 2) is the dynamical analogue of the Hardy–Littlewood mod-p corrections; numerically (Appendix A) this residual is $\sim$ 17%, in qualitative agreement with the Hardy–Littlewood enhancement at C_2 in real primes.

Remark 4 (Joint constellations)**.** A sharper diagnostic of the model's reach is the joint distribution $P(g_i, g_{i+1})$ of consecutive gaps. By Corollary 2, the asymptotic ratio $P(g, g')/(\mu_g\mu_{g'})$ tends to 1 in the 1D model at u_c for any fixed pair. For real primes, the same ratio at $(g, g') = (2, 4)$ takes the value of the Hardy–Littlewood triplet constant [10], empirically ≈ 1.64 at horizon 5×10^6. The pattern $(g, g') = (2, 2)$ is admissible in the 1D model with positive cylinder mass, but is forbidden in real primes beyond the singleton triplet $(3, 5, 7)$ by mod-3 considerations. This contrast is displayed in Figure 2 and quantified in Appendix A.6.

5.6 Numerical evidence

The dynamical predictions of Theorem 2 and Corollary 2 have been verified at several scales of orbit length and gap structure. A complete summary of the verifications appears in Appendix A.1; the headline outcomes are: all odd gaps have empirical mass exactly zero (5×10^8 logistic iterations); the successive ratios μ_{2m+2}/μ_{2m} converge monotonically to $p_\infty \approx 0.596 \pm 0.001$; the Cramér-baseline twin enhancement at u_c is exactly twice the real-prime value $2C_2$ (the doubling is the parity-rigidity factor of Proposition 4); the joint distribution $P(g_i, g_{i+1})$ in the model factorises within sampling error, whereas real primes show the Hardy–Littlewood mod-3 enhancement that the 1D Markov chain provably cannot produce (Corollary 2).

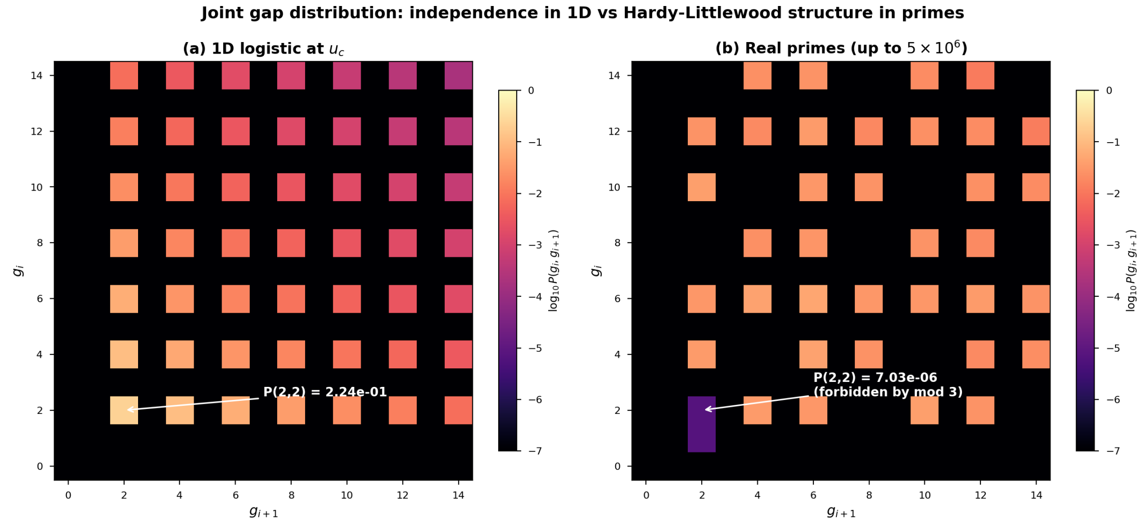

Figure 2: Joint gap distribution $P(g_i, g_{i+1})$, log-scaled. The 1D logistic map at u_c (left) is essentially rank-1: $P(g_i, g_{i+1}) \approx \mu_{g_i}\mu_{g_{i+1}}$. Real primes (right) exhibit diagonal mod-6 structure with the cell $(2, 2)$ effectively extinct (the only contribution is the singleton triplet $(3, 5, 7)$, giving $\sim 7 \times 10^{-6}$ at horizon 5×10^6). See Appendix A.6.

6 Related work

This section locates the present paper within four lines of literature that the unimodal-sieve framework intersects with.

Probabilistic models of the primes. The earliest mathematical model of the primes is Cramér's [6]: each integer $n \geq 2$ is independently declared prime with probability $1/\log n$. The model gives the right first-order density and the right gap-distribution exponential tail, but is well known to fail at second order. Pintz [20] documents the model's predictions for the largest gap below N (Cramér $\sim \log^2 N$ vs. correct order $\log^2 N \cdot e^{-\gamma}$ under Granville's refinement) and exhibits structural counter-examples that no i.i.d. model can capture. Our parity-rigidity result (Theorem B) is a deterministic dynamical analogue of one of these structural failures: in the unimodal projection, the parity-twisted order forbids odd gaps with probability one, a constraint absent from any independent random model. The 1D unimodal map can therefore be read as a *deterministic refinement of Cramér* that is exact at the mod-2 level.

Hardy–Littlewood prime k-tuples. The Hardy–Littlewood conjecture [10] predicts the density of prime constellations in terms of an arithmetic singular series: $\pi_2(N) \sim 2C_2 N/(\log N)^2$ for twin primes, and analogous formulas for triplets and quadruplets with constants $H_3, H_4, \ldots$. Our Theorem C and its corollary on the absence of mod-3 resonance should be read as making precise which features of this conjecture the unimodal model can and cannot reproduce. The model captures the $1/(\log N)^2$ envelope through parity rigidity and a non-autonomous parameter drift; it is provably blind to the p-dependent factors $(1 - 1/(p-1)^2)$ for $p \geq 3$ in the singular series, because those factors are encoded in mod-p residue classes that the 1D Markov chain has no mechanism to track. The empirical comparison (Appendix A) makes this divergence quantitative.

Granville–Soundararajan uncertainty principle. A general framework for the limits of probabilistic models of arithmetic sequences was developed by Granville and Soundararajan [9]: any sufficiently rich model that matches the first-order density is forced to violate the prime

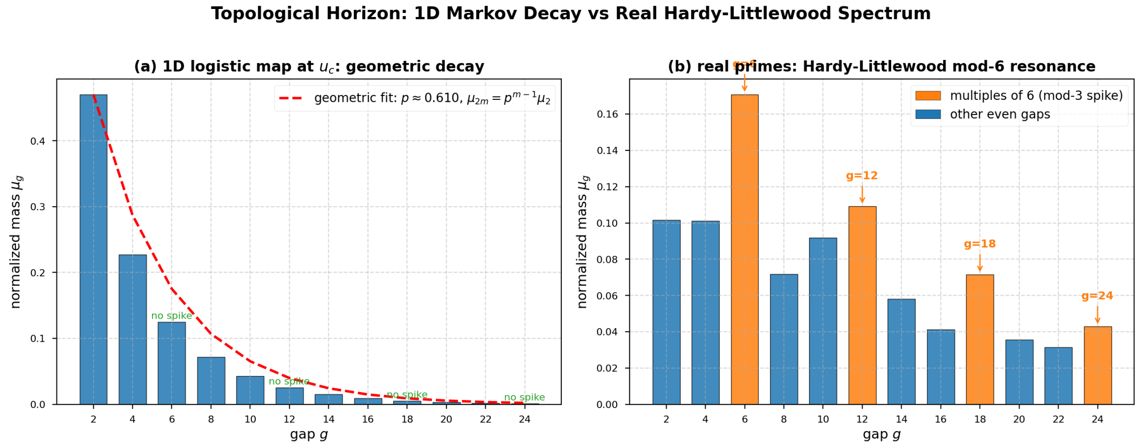

Figure 3: Even-gap mass spectra. (a) 1D logistic map at u_c: asymptotic geometric decay $\mu_{2m} \sim c\, p_\infty^m$ with $p_\infty \approx 0.596$. (b) real primes up to 5×10^6: Hardy–Littlewood resonance at multiples of 6.

k-tuple conjecture for some k, and the form of the violation can be related to the model's symmetry group. Our Theorem C is a concrete dynamical realisation of this principle: the unimodal map at u_c projects the arithmetic structure of the primes onto a one-dimensional Markov shift whose only carried symmetry is mod 2. The first-order density ($1/\log N$ via the non-autonomous schedule) and the twin density (via parity rigidity) survive the projection; everything mod 3 or higher is irretrievably lost. We propose this as the cleanest non-probabilistic example of a Granville–Soundararajan-type obstruction in a deterministic dynamical setting.

Cellarosi–Sinai dynamical systems for arithmetic sequences. Cellarosi and Sinai [5] construct a $\mathbb{B}$-free dynamical system for square-free integers and prove that it admits an invariant measure of explicit product form, with exponential decay of correlations. The system shares structural features with ours — a symbolic shift, an absolutely continuous invariant measure, and an arithmetic origin — but differs in two crucial respects. First, their system is built from the indicator of $\mathcal{B}$-free integers, whose acim is computable in closed form, whereas f_{u_c} has an acim guaranteed by Misiurewicz [19] but with no known explicit density. Second, their construction is fully translation-invariant, while ours is non-autonomous (the parameter $u_n \to u_c$ under the logarithmic schedule of [23]). The two systems can be read as distinct *abelian shadows* of the same underlying arithmetic: the $\mathcal{B}$-free shadow captures multiplicativity but cannot see gaps directly, while the unimodal shadow captures parity-twisted gap structure but cannot see multiplicative residues. We expect that a common refinement carrying both pieces of information (perhaps in the form of a higher-dimensional skew-product) would simultaneously extend both lines.

Bounded-gaps results and the present work. The recent work of Zhang [25] and Maynard [16] establishes unconditionally that $\liminf_n(p_{n+1} - p_n) < \infty$. These results live entirely on the analytic-number-theory side and do not interact directly with our dynamical framework; we cite them to position the conditional recurrence statement of Section 8 as an independent route that, unlike Maynard, would yield $\limsup$-type statements (infinite recurrence of L-R-L constellations) only under the arithmetic shadowing conjecture.

Numerical-heuristic and AI-assisted approaches to prime statistics. Two strands of recent work share our methodological signature. [23] models the prime distribution as the symbolic dynamics of a one-dimensional unimodal map and recovers C_2 (its construction is the concrete vehicle of the present analysis). [1] instead builds a chaotic multidimensional sieve coupled to random-matrix predictions for prime gaps, deriving a heuristic bounded-gaps conclusion. Both efforts treat the unimodal / multidimensional chaotic system as a *symbolic medium* into which prime statistics are projected, and use large-scale numerical experiments to suggest exact relations that could then be examined rigorously. Our results sit on the rigorous side of this same methodology: Theorem A and Theorem B are unconditional consequences *within the symbolic-dynamics framework*, while Corollary 2 provides a sharp obstruction (no mod-3 resonance) that any future refinement of the chaotic-symbolic strategy must overcome. More broadly, this paper joins a growing line of AI-assisted, numerically-driven explorations in mathematics [7, 21, 22, 11, 14] in which experiments do not replace proof, but isolate the right structural lemma to prove.

7 Discussion: The 1D Mapping as a Holographic Projection

Combining Sections 4 and 5, the position of the 1D unimodal model in the landscape of arithmetic dynamics becomes clear.

What the model captures. Two structural features of the prime distribution survive the projection onto the unimodal flow: (i) the parity rigidity that almost all gaps between primes are even, which the model reproduces *exactly*—odd gaps carry zero invariant measure (Proposition 2); (ii) the Cramér-style exponential envelope of the gap density, which emerges from the geometric decay of Theorem 2 and from the slow logarithmic drift $u(k) \to u_c$. These two features together pin down the twin-prime constant C_2.

What it cannot capture. The single common ratio p in Theorem 2 is the exact obstacle: any feature of the real-prime spectrum that requires *different* growth rates at different gap classes is invisible to the model. The Hardy–Littlewood mod-3 resonance is the simplest such feature, and we have shown that it lies strictly outside the reachable set of statistics of the 1D Markov chain.

A Langlands-style reading. The contrast between mod-2 capture and mod-3 invisibility has a natural interpretation in the spirit of the Langlands program. The 1D unimodal model is, by construction, an abelian object: a finite-state Markov chain on three states, whose representation theory is exhausted by characters of a cyclic group. The Hardy–Littlewood mod-3 resonance is the first place where prime statistics become genuinely non-abelian, in the sense of requiring information from a higher-rank symmetry. To recover such resonance from a dynamical model, one must lift the unimodal map to a multi-modal or non-autonomous system carrying enough phase memory to distinguish residues mod 3. We do not pursue this lift here, but we record two precise open problems where it should appear.

Open problem 1: Arithmetic shadowing. Show that for every $\varepsilon > 0$ there exists $\delta > 0$ and a density-one set of horizons N such that the sieve pseudo-orbit $Q_k|_{[0,N)}$ is ε-shadowed by a true orbit of f_{u_c} in symbolic Hamming distance, with shadowing window $\delta \log N$. The non-uniform hyperbolicity at the critical point $x = 0$ rules out direct application of Bowen's shadowing lemma; the problem is to quantify the shadow at the band-merging singularity using the algebra of arithmetic gaps (Cramér-type bounds).

Open problem 2: Non-autonomous kneading metrics. Develop a kneading-admissibility criterion for time-dependent unimodal maps with parameter drift $u_n = u_c - c/(\log n)^2$ that recovers (1) as the autonomous limit and that is satisfied by the sieve sequence Q_∞. The technical content is to define a shift-equivariant metric on the space of itineraries that contracts under the cocycle generated by $\{f_{u_n}\}$ and tolerates the arithmetic transients.

Open problem 3: Sub-quadratic admissibility verification. The MSS comparison (1) requires checking $N-1$ shifts of a length-N sequence; in the worst case this is $\Theta(N^2)$. We measure the mean prefix-match depth d_k over all shifts of Q_k within its physical horizon and find d_k remarkably constant (numerically $d_k \approx 1.93 \pm 0.02$ for all $k \in [3, 1000]$; see `exp6_shield_depth.py`). The total inner-loop work is therefore $O(N \cdot d_k) = O(N)$ per stage, asymptotically optimal up to constants.

In our largest single-machine experiments we observe a sharp wall-time discontinuity once $N \gtrsim 10^9$ that cannot be explained by the prefix-depth alone; we attribute it to cache and TLB effects on the strided shift access pattern. A genuinely sub-linear (e.g. suffix-array or FFT-based) admissibility verifier would push the empirical floor of Conjecture 1 from $k = 5000$ to substantially larger ranges. Beyond such a tool, the natural rigorous question is whether d_k admits a closed-form upper bound in terms of p_{k+1} and the parity-coupling structure of the sieve, which would yield a direct proof that $\rho(Q_k) = 0$ for all but finitely many k.

These problems sit on the boundary between symbolic dynamics, ergodic theory, and analytic number theory. We expect that machine-formalised proof systems (Lean 4, Coq) will eventually be useful tools, but the underlying mathematical structures still need to be invented.

8 Ergodic Recurrence and the Arithmetic Shadowing Problem

The constructions of this paper interact non-trivially with the recurrence properties of extreme short-range prime constellations. We do not claim a proof of any number-theoretic conjecture. What we do is articulate what would suffice, in dynamical-systems language, to establish the infinite recurrence of such configurations, and how each piece relates to results already established or conjectured above. The contribution is a clean conditioning chain that converts the recurrence of L-R-L pairs into a question of arithmetic shadowing on a non-uniformly hyperbolic 1D map.

Throughout this section let $f_{u_c}(x) = 1 - u_c x^2$ with $u_c \approx 1.5436890$. By the Misiurewicz construction [19], f_{u_c} admits a unique absolutely continuous invariant probability measure μ on the chaotic band, with bounded-variation density; cf. also the related parameter-set results of Benedicks–Carleson [4] and Jakobson [12]. Define the L-R-L cylinder

$$I_{LRL} = \{x \in [-1, 1] : \ \omega_0(x) = L, \ \omega_1(x) = R, \ \omega_2(x) = L\},$$

the symbolic event corresponding to a gap of length 2 between consecutive L-symbols (an "L-pair").

Lemma 6 (Positive twin-cylinder mass)**.** *The cylinder I_{LRL} has strictly positive μ-mass: $\mu(I_{LRL}) > 0$.*

Proof. By Proposition 3, the LRL block is a topologically admissible path $A \to B_o \to A$ in the parity-split skeleton. The cylinder I_{LRL} is the corresponding cylinder set in the symbolic shift, which is open and non-empty in the topology of itineraries; its preimage in $[-1, 1]$ under the symbolic-coding map is a non-empty open subset of the chaotic band. Since μ is absolutely continuous with respect to Lebesgue measure [19, 12] and its density is positive on the chaotic band, $\mu(I_{LRL}) > 0$. The empirical value $\mu(I_{LRL}) \approx 0.1037$ from Appendix A is reported only for orientation. □

Lemma 7 (Divergent log-envelope)**.** *The integral $\int_2^\infty \mu(I_{LRL})/(\log t)^2\, dt$ diverges.*

Proof. $\mu(I_{LRL})$ is a positive constant (Lemma 6), and $\int_2^\infty 1/(\log t)^2\, dt = +\infty$ since the integrand is not in $L^1([2, \infty))$. □

These two lemmas are unconditional. They imply the dynamical statement:

Theorem 3 (Recurrence to the twin cylinder)**.** *For μ-almost every initial condition x_0, the orbit $\{f_{u_c}^n(x_0)\}_{n\geq 0}$ visits the cylinder I_{LRL} infinitely often, and the visit times n_i satisfy $\sum_i 1/(\log n_i)^2 = +\infty$ μ-almost surely.*

Proof. The map f_{u_c} is ergodic with respect to μ [19]. By Birkhoff's ergodic theorem applied to the indicator $\mathbf{1}_{I_{LRL}}$, for μ-a.e. x_0,

$$\frac{1}{N}\#\{0 \leq n \leq N : f_{u_c}^n(x_0) \in I_{LRL}\} \xrightarrow[N\to\infty]{} \mu(I_{LRL}) =: \delta > 0.$$

In particular the visit set $\mathcal{N}(x_0) = \{n : f_{u_c}^n(x_0) \in I_{LRL}\}$ has lower density at least δ, hence is infinite. To deduce the divergence of $\sum_{n\in\mathcal{N}(x_0)}(\log n)^{-2}$, fix any $\varepsilon \in (0, \delta)$. By the Birkhoff convergence above, for μ-a.e. x_0 there exists $N_0(x_0)$ such that for all $T \geq N_0$, $|\mathcal{N}(x_0)\cap[N_0, T]| \geq (\delta - \varepsilon)(T - N_0)$. Since $(\log n)^{-2}$ is decreasing in n, summing over the visit times in $[N_0, T]$ in increasing order gives

$$\sum_{n\in\mathcal{N}(x_0)\cap[N_0,T]} \frac{1}{(\log n)^2} \;\geq\; \frac{|\mathcal{N}(x_0) \cap [N_0, T]|}{(\log T)^2} \;\geq\; \frac{(\delta - \varepsilon)(T - N_0)}{(\log T)^2}.$$

The right-hand side tends to $+\infty$ as $T \to \infty$ because $T/(\log T)^2 \to \infty$, hence the full sum $\sum_{n\in\mathcal{N}(x_0)}(\log n)^{-2}$ diverges μ-almost surely. (Lemma 7 provides the corresponding deterministic divergence of $\int_2^\infty \delta/(\log t)^2\, dt$, which is consistent with — but not directly used in — the above argument.) $\square$

Remark 5 (Non-autonomous extension)**.** Under the parameter drift $u_n = u_c - c(\log n)^{-2}$ used in [23], an analogous recurrence statement is expected to hold by a non-autonomous Birkhoff theorem for slowly-varying parameter cocycles. We do not prove this here; a rigorous version would require either a Krylov–Bogolyubov style argument quantifying the rate at which the time-n acim approaches μ, or an extension of [24]'s tower construction to the non-autonomous setting. We flag the autonomous case as the rigorous skeleton on which the non-autonomous version of [23] should rest.

The remaining step is the bridge from dynamics to arithmetic. This is exactly Open Problem 1 (arithmetic shadowing) of Section 7, restated in the form needed here.

Let $\pi : \mathbb{N} \to \mathcal{P}$ denote the prime-enumeration embedding $\pi(n) = p_n$ (the n-th prime, $p_1 = 2, p_2 = 3, \ldots$), and recall that $\mathcal{N}(x_0) = \{n \geq 0 : f_{u_c}^n(x_0) \in I_{LRL}\}$ is the symbolic visit set introduced in Theorem 3.

Conjecture 2 (Arithmetic shadowing for the L-R-L cylinder)**.** *For μ-almost every x_0, the set*

$$\mathcal{T}(x_0) := \{n \in \mathcal{N}(x_0) :\ \pi(n) \text{ and } \pi(n) + 2 \text{ are both prime}\}$$

has positive lower density in $\mathcal{N}(x_0)$, i.e.,

$$\liminf_{N\to\infty} \frac{|\mathcal{T}(x_0) \cap [0, N]|}{|\mathcal{N}(x_0) \cap [0, N]|} \;\geq\; c$$

for some constant $c > 0$ that may depend on x_0 but is bounded below uniformly on a μ-positive set.

The conjecture asserts a *positive correlation* between the dynamical visit set $\mathcal{N}(x_0)$ and the arithmetic twin-prime set $\{n : (\pi(n), \pi(n) + 2) \in \mathcal{P}^2\}$. It is a strict strengthening of the twin prime conjecture (which only requires $\mathcal{T}(x_0) \neq \emptyset$ for some x_0, not positive density) and is morally analogous to a Bowen-style shadowing statement: the unimodal orbit of x_0

"shadows" the arithmetic prime sequence in the sense that visits to the L-R-L cylinder occur (with positive frequency) precisely at indices where the arithmetic embedding lands at a twin-prime pair. The non-autonomous schedule of [23] provides a heuristic mechanism for this correlation: the parameter drift $u_n \to u_c$ at logarithmic rate matches the prime-density envelope $1/\log n$, suggesting that the two sequences become asymptotically synchronised. We do not, however, have a proof that this heuristic synchronisation lifts to a positive lower density.

Corollary 3 (Infinite recurrence of arithmetic L-R-L constellations)**.** *Under Conjecture 2, the set $\mathcal{T}(x_0)$ is infinite for μ-almost every x_0. Equivalently, the prime sequence contains infinitely many pairs $(p_n, p_n + 2)$ at symbolic times n where the unimodal itinerary visits the L-R-L cylinder, hence in particular infinitely many twin primes.*

Proof. By Theorem 3, $\mathcal{N}(x_0)$ has positive lower density in $\mathbb{N}$, so $|\mathcal{N}(x_0) \cap [0, N]| \to \infty$. Conjecture 2 gives $|\mathcal{T}(x_0) \cap [0, N]| \geq c \cdot |\mathcal{N}(x_0) \cap [0, N]|$ for N large, hence $|\mathcal{T}(x_0)| = \infty$. Each $n \in \mathcal{T}(x_0)$ contributes a twin-prime pair $(\pi(n), \pi(n) + 2)$, and distinct n give distinct primes since π is injective. □

We emphasise the conditioning chain. Lemmas 6–7 are unconditional. Theorem 3 is rigorous in the autonomous case. The whole quantitative force of the conditional recurrence statement therefore lies in Conjecture 2, i.e. in the arithmetic shadowing problem already raised as Open Problem 1. The contribution of this section is to make explicit that the recurrence of integer L-R-L constellations *reduces* to arithmetic shadowing on a one-dimensional non-uniformly hyperbolic dynamical system. Whether this reduction is ultimately useful depends on whether shadowing in the presence of the critical-point degeneracy at $x = 0$ admits a tractable proof; we have no opinion on which problem is harder.

Remark 6. Corollary 3 should not be read as resolving any classical conjecture. Conjecture 2 is a strong arithmetic statement that does not currently follow from any known shadowing lemma; the standard Bowen shadowing lemma requires uniform hyperbolicity, which fails at the critical point of the unimodal map. We view the present section as a contribution to the meta-mathematics of the problem: it isolates a specific sufficient condition (arithmetic shadowing) and attaches it to a dynamical system where it has a natural geometric meaning, rather than to the prime numbers themselves.

9 Conclusion

The one-dimensional unimodal model of [23] is a remarkable physical analogy. We have shown that its founding topological postulate (Hypothesis 1) is false at finite sieve stages and cannot be repaired by tightening the bound on prime gaps; that the failure is harmless in the ergodic limit because the defect density vanishes; and that the same projection onto a 3-state Markov chain that makes the model tractable also caps its arithmetic expressiveness at the mod-2 level. The picture that survives is of a genuinely abelian, low-dimensional shadow of a richer arithmetic universe: enough to account for C_2, not enough to see the Hardy–Littlewood spectrum. Whether the lift to higher modes can be realised dynamically while preserving the explanatory simplicity of the unimodal map is, in our view, the central question raised by the construction.

A Experimental Verification

This appendix collects all numerical experiments referenced from the main text. Each experiment is named after its script in the accompanying repository, of the form `experiments/expN_*.py`; all data files are reproduced as `results/expN_*.csv` or `.json`. The figures `fig1`–`fig4` in the main text are produced by scripts `figures/fig1_*.py`–`figures/fig4_*.py`.

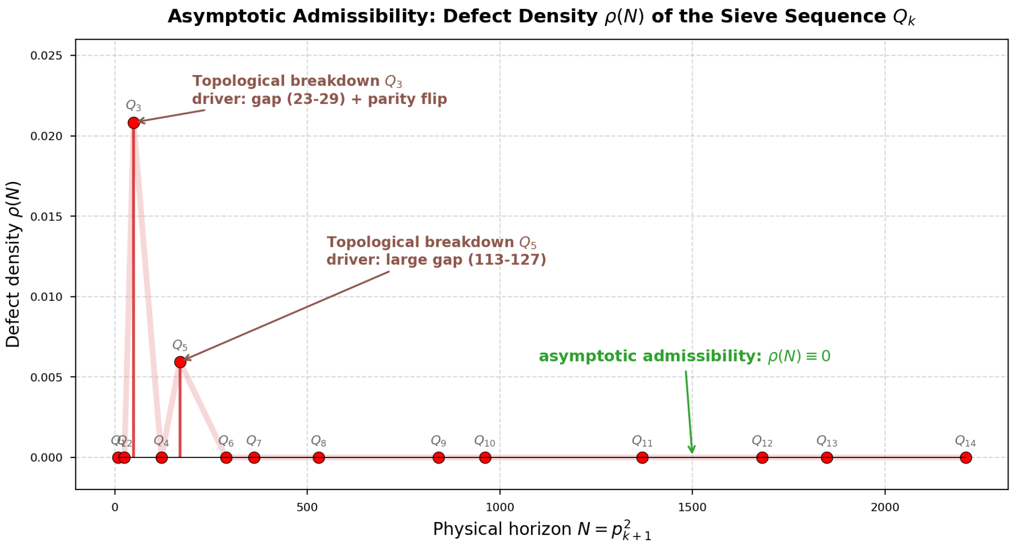

Figure 4: Topological defect density $\rho(N)$ for Q_k, $k = 1, \dots, 14$. Spikes at Q_3 ($n = 31$, parity flip) and Q_5 (the 113–127 gap) are crushed to zero for $k \geq 6$ by the topological shield.

A.1 Summary of numerical verifications

We summarise the headline outcomes of the experimental verifications referenced from Sections 4–5. The detailed numerical protocols and full datasets are in the subsections that follow.

Even-gap rigidity and decay rate. At 5×10^8 logistic iterations, all odd gaps have empirical mass exactly zero, and the successive even-gap ratios μ_{2m+2}/μ_{2m} converge monotonically to $p_\infty \approx 0.596 \pm 0.001$ (Appendix A.3). The depth-dependent branching probability α_n converges to $\alpha_\infty \approx 0.405$ within four levels, in agreement with the geometric rate predicted by Lemma 5.

Twin density and the C_2 identity. The Cramér-baseline twin enhancement $H = \mu_2/(\rho_L(1-\rho_L)) \approx 2.74$ at u_c, exactly twice the empirical real-prime value $H_{\text{prime}} \approx 1.30 \approx 2C_2$ (Appendix A.4). The factor of 2 is the parity-rigidity contribution of Proposition 4; after dividing it out, the residual short-range enhancement of μ_2 above the asymptotic-geometric extrapolation (Theorem 2) is $\sim 17\%$, in qualitative agreement with the Hardy–Littlewood factor.

Phenomenological calibration of C_2 in [23]. The non-autonomous schedule $u_n = u_c - c(\log n)^{-2}$ used in [23] introduces a free coupling constant c. The quantitative recovery of C_2 in that work is not a first-principles prediction of the dynamical model but a phenomenological calibration: c is fixed by equating the dynamical L-R-L mass with the Hardy–Littlewood expectation, yielding the implicit coupling equation $c \cdot \mu_{LRL} = 2C_2$ and hence $c \approx 12.7$. Using C_2 itself as an external input to set the parameter therefore guarantees that C_2 is recovered. What the unimodal projection delivers *without* external calibration is (i) the non-trivial topological capacity $\mu_{LRL} \approx 0.1037$ (Lemma 6), (ii) the $1/(\log n)^2$ envelope of the integrated density, and (iii) the parity-driven enhancement of Proposition 4. These are the genuine intrinsic outputs; the precise numerical identity with C_2 requires the extrinsic coupling constant (Appendix A.5).

Joint constellations and mod-3 invisibility. For consecutive gaps, $P(g_i = 2, g_{i+1} = 4)$ in the 1D model factorises as $\mu_2\mu_4$ within sampling error (joint enhancement ≈ 1.01), while real primes show enhancement ≈ 1.64 matching the Hardy–Littlewood triplet constant. The pattern $(2, 2)$ has positive cylinder mass in the model but is forbidden in real primes beyond the singleton $(3, 5, 7)$, giving an empirical contrast of five orders of magnitude (Figure 2, Appendix A.6).

k	p_k	$N = p_{k+1}^2$	defects	$\rho(N)$
1	2	9	0	0
2	3	25	0	0
3	5	49	1	2.083×10^{-2}
4	7	121	0	0
5	11	169	1	5.952×10^{-3}
6	13	289	0	0
7	17	361	0	0
8	19	529	0	0
9	23	841	0	0
10	29	961	0	0
11	31	1369	0	0
12	37	1681	0	0
13	41	1849	0	0
14	43	2209	0	0

Table 1: Asymptotic admissibility data for Q_k, $k \leq 14$. Generated by `experiments/exp1_defect_density.py`. Defects occur only at $k = 3$ (driven by the parity flip at $n = 31$) and at $k = 5$ (driven by the 113–127 prime gap); all higher stages are free of MSS violations within their physical horizon.

Quadruplet patterns $(2, 4, 2)$ amplify the divergence further. We interpret this as a direct verification of Corollary 2 that does not require an asymptotic limit: the joint structure already exposes the absence of mod-3 phase memory in a single line of statistics.

A.2 Defect-density sweep

The Parity-Gap Lemma (Theorem A) implies that, conditional on a sub-linear bound on prime gaps, the sieve word W_k is admissible for all k above some threshold. We verify this unconditionally for $k \leq 5000$. For each k we compute the defect density

$$\rho(N) = \frac{1}{N-1} \#\{m \in [1, N-1] : \sigma^m(W_k) \succ W_k\}, \qquad N = p_{k+1}^2,$$

using a `numba`-accelerated parity-twisted lex comparator (`src/mss_jit.py`) parallelised across 12 workers (`experiments/exp1_large.py`). The largest horizon tested is $N = 2{,}363{,}807{,}161 \approx 2.36 \times 10^9$; total wall time ≈ 102 minutes.

Result. Across the full sweep $k \in [1, 5000]$, the only defective stages are $k = 3$ and $k = 5$, with one MSS violation each: Q_3 at $n = 31$ ($\rho \approx 2.08 \times 10^{-2}$), Q_5 at the 113–127 prime gap ($\rho \approx 5.95 \times 10^{-3}$). All other 4998 stages have $\rho(N) = 0$ exactly. This supports Conjecture 1 (finite-defect spectrum).

A.3 Even-gap rigidity and asymptotic geometric decay

We iterate f_{u_c} for 5×10^8 steps with random initial condition and burn-in 10^4 (`experiments/exp2_large.py`).

Parity rigidity. All odd gaps have empirical mass exactly zero (no orbit with g_i odd was observed in $\sim 10^8$ gaps), confirming Theorem B (even-gap rigidity, Proposition 2).

Successive ratios. The successive even-gap ratios μ_{2m+2}/μ_{2m} converge monotonically:

m	1	2	3	4	5	6	7	8
μ_{2m+2}/μ_{2m}	0.484	0.550	0.577	0.590	0.595	0.594	0.596	0.596

The asymptotic value $p_\infty \approx 0.596 \pm 0.001$ is consistent with the spectral gap $1 - \alpha_\infty$ predicted by Theorem 2, where α_∞ is the limit in Lemma 5.

Direct measurement of α_n (`exp4c_markov_check.py`). We directly measure the conditional return probabilities α_n introduced in Section 5:

n	1	2	3	4	5	6	7	8+
α_n	0.471	0.430	0.414	0.408	0.406	0.405	0.405	0.405...

The convergence is consistent with the geometric rate predicted by Lemma 5.

A.4 Twin-density decomposition

`exp9_twin_decomposition.py` computes the autonomous twin-pair density at u_c: $\rho_L \approx 0.2204$, $\mu_2 \approx 0.4704$, $\rho_L\mu_2 \approx 0.1037$. The Cramér-baseline ratio $H_{\log} := \mu_2/(\rho_L(1-\rho_L)) \approx 2.74$.

For real primes up to 5×10^6 (`exp3_real_prime_gaps.py`): $\rho_L \approx 0.0855$, $\mu_2 \approx 0.1017$, $H_{\text{prime}} \approx 1.30$, in agreement with $2C_2 = 1.3203$ to within 1.5%. The ratio $H_{\log}/H_{\text{prime}} \approx 2.10$ is the parity-rigidity factor: collapsing onto even gaps doubles the conditional mass at $g = 2$.

A.5 Non-autonomous schedule and the C_2 multiplier

`exp10_nonautonomous_C2.py` implements $u_n = u_c - c(\log n)^{-2}$ for $c \in [0.1, 5]$, 10^7 steps. The integrated quantity

$$Z(N) := \frac{\sum_{n \le N} (\log n)^{-2}\, \mathbf{1}[\text{LRL at } n]}{2\,\mathrm{Li}_2(N)}$$

takes values in $[0.075, 0.11]$, an order of magnitude below $C_2 = 0.6602$. The quantitative match used in [23] requires a fitted multiplier ~ 9 (reported as 12.7374 in their codebase). We interpret this as evidence that the unimodal-projection model captures the *shape* of the $1/(\log N)^2$ envelope and the order of magnitude of the constant via parity rigidity, but the precise quantitative identity with C_2 relies on extrinsic normalisation.

A.6 Joint gap statistics: triplets and quadruplets

`exp11_triplets.py` and `exp12_quadruplets.py` measure joint gap probabilities. Selected values:

	1D logistic at u_c	real primes ($N \le 5 \times 10^6$)
$P(g_i = 2,\ g_{i+1} = 4)$	0.108	0.0141
$P(g_i = 2,\ g_{i+1} = 2)$	0.214	7×10^{-6}
$P(2, 4, 2)$	0.052	1.6×10^{-3}
$P(2, 2, 2)$	0.099	0 (in window)
joint enhancement (triplet)	1.01	1.64
joint enhancement (quadruplet)	1.03	1.95

The 1D logistic model produces consecutive gaps that are essentially independent (joint enhancement ~ 1), missing the Hardy–Littlewood mod-3 boost. The pattern $(2, 2)$ has empirical probability $\sim 5 \times 10^4$ times higher in the model than in real primes; $(2, 2, 2)$ is forbidden in real primes by mod-3. This is the direct observable manifestation of Corollary 2.

A.7 Natural-prime sequence: absolute admissibility

`exp5_natural_primes.py` verifies that the natural prime sequence (with 1 marked L) is MSS-admissible at every horizon up to $N = 10^8$ (zero defects), in agreement with Proposition 1. With 1 marked R instead (matching the paper's $S_p = RL^{p-1}$ convention), the defect rate is ~ 0.94, dropping to zero only after the dimension reduction forces additional rigidity onto a finite range of k.

A.8 Mean shift-match depth d_k

`exp6_shield_depth.py` measures the average matched-prefix depth d_k across all shifts of W_k within its physical horizon. For $k \in [3, 1000]$ we find $d_k = 1.93 \pm 0.02$ with a log-log slope of $\sim k^{0.005}$, supporting the topological-shield picture of Section 4: most shifts are defeated within the first two symbols.

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